Fabric Side to Side Shade Variation Handling Guideline

布料橫向色差處理策略指引

Purpose and Scope / 目的與適用範圍

This guideline outlines the inspection, grouping, cutting, and post -treatment strategies to manage color variations across the width (side to side shading) of polyester + elastane knitted fabrics used for T -shirts. It aims to ensure visual consistency, minimize fabric waste and reduce customer complaints. 本指引旨在 說明當針織布料（ Polyester + Elastane ）於布幅左右或中間出現顏色差異時, 成衣廠可採取的檢驗, 分組, 裁剪與後處理策略. 適用於 T-shirt 類產品之布料檢驗與生產流程, 以確保成衣整體外觀一致性, 並降低報廢與客訴風險.

Definition

"Side to side shade variation" refers to visible color differences between the left, center, and right sections of the same fabric roll, often caused by uneven dyeing tension, dye uptake, pressure, or fabric orientation during processing. "橫向色差" 指布料在同一布幅的左, 中, 右位置呈現不同色階或亮度, 常因染整張力, 上染速率, 壓輥壓力或布紋方向差異所造成.

Principle

Follow the five step approach: 1. Identify -> 2. Group -> 3. Cut -> 4. Utilize -> 5. Control.

以 1. 識別 -> 2. 分組 -> 3. 裁剪 -> 4. 利用 -> 5. 控管 五步驟進行.

Procedures

Step 1. Fabric Inspection & Shade Band Setup

- Before cutting, QC should take 10 x 10 cm swatches from the left, center, and right
- sections. 於裁剪前, QC 需於布幅左, 中, 右各取一片 10 x 10 cm 布樣
- Compare them against the standard to create a Shade Band (A/B/C). 將三段顏色與基準色比對，建立 Shade Band （色階帶）例如 A/B/C 級.
- Mark each section with stickers or color labels. 依顏色深淺標註於布邊.

Step 2. Shade Grouping/分色分組

- Classify fabric into shade groups. 依檢驗結果將布料分為不同 A/B/C.
- Avoid mixing different shade groups. 避免不同 Shade Group 混裁.
- Assign shades by size if necessary. 可依布量分配不同尺寸生 產.

Step 3. Cutting & Layout Planning/裁剪規劃與排版

- Cut according to shade marks to prevent mismatch. 裁剪員應依 Shade 標示分區裁剪，避免單件衣服左右布色差.
- Front/back panels from same zone. 前後身片應取自同一區.
- Use symmetrical sleeve cutting. 袖子採對稱裁剪策略.
- Avoid mixing rolls. 避免跨卷裁剪.

Step 4. Creative Utilization/顏色差異的創意利用

- Use off -shade areas for neck rib, cuff, binding, neck tape, or decorative panels. 可將色差區用於領口, 袖口, 包邊, 內領條 or 裝飾區.

Step 5. Post -Treatment & Inspection/成衣後處理與檢驗

- Apply garment/enzyme wash for mild variation. 對輕微色差批次可進行 Garment Wash 或 Enzyme Wash.
- Sort garments before shipment. 出貨前進行 Shade Sorting , 確保同箱一致.
- Indicate Shade Code on packaging. Shade Code 應標於包裝與文件

Shade Group vs Size Example/Shade 分組與尺寸對應範例

Shade Group	色階說明	建議尺寸	裁剪區域建議
A	偏淺 / Light Tone	S / XS	布左側 / Left Side
B	中間 / Medium Tone	M	布中央 / Center
C	偏深 / Dark Tone	L / XL	布右側 / Right Side

Remarks: Quality Highlight/品質控管重點

All fabrics must pass shade band approval. 所有布料裁前須通過 Shade Band 認可.

No garment should mix different shade groups. 同一件衣服不可混用不同 Shade Group.

Off-shade trims require Brand's approval. 領口, 袖口使用異色前須與品牌確認.

Maintain a Shade Control Log. 建立 Shade Control Log 追蹤批次.